George Paul

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My research path stems from an MSc dissertation that grew into a methodological paper on Bayesian shrinkage priors. It continued into an applied research role in industry, where I moved into fine-tuning and evaluating vision-language and language models alongside my team, work that became a peer-reviewed EC3 2026 paper. That transition gave me hands-on experience of the gap between how models are evaluated on small held-out sets and how they behave in deployment. I am looking for a CDT programme where there is genuine opportunity to collaborate with and learn from a cohort.

Publications & Research

The Adaptive Elastic Horseshoe Prior — Feb. 2025 – Present

First author, with Prof. Michael Tipping — manuscript prepared for submission to the Electronic Journal of Statistics — *University of Bath*

- **Problem**: Shrinkage priors fix their regularisation geometry in advance, so no existing method adapts when sparse and correlated structures appear in the same data. Building energy data, where I first met the problem, is exactly that: a few strong drivers alongside many weak, collinear ones.
- **Approach**: A prior that learns the sparsity/grouping balance from the data during inference. Showed the adaptive updates are approximate M-steps of an EM algorithm on a variational lower bound, so the procedure is principled rather than heuristic, and established minimax-optimal posterior contraction in the sparse normal-means setting, matching the horseshoe rate.
- **Evaluation**: Applied within an ARD framework to energy-use-intensity prediction with full uncertainty decomposition, then generalised post-degree across eight simulation studies and a standard benchmark dataset, mapping where the method helps and where it does not. Convergence failures and calibration limits reported clearly.
- **Open questions**: Extending the contraction theory to correlated designs, improving predictive calibration in dense high-dimensional designs.
- Began as the MSc dissertation (81%); developed further independently alongside full-time employment into a first-author manuscript.

Benchmarking Deep Learning for Engineering Drawing Layout Detection — 2025 – 2026

Co-author, peer-reviewed at EC3 2026 (DOI: 10.35490/EC3.2026.305) — contribution: vision-language model methodology & evaluation — *Buro Happold*

- **Problem**: Document layout models are trained and benchmarked almost exclusively on text-centric corpora such as invoices and academic papers. Their reported performance therefore says little about behaviour on engineering drawings, which differ in essentially every respect that matters: sparse line-based geometry rather than dense text, extreme aspect ratios, and structure carried by spatial relationships rather than reading order. No prior work had quantified that gap.
- **Approach**: Worked with facade engineers to agree what should be labelled and to build an annotated dataset of real drawings, then ran a controlled comparison of five architectures. I designed the evaluation protocol and led the vision-language model arm, fine-tuning Qwen3-VL via parameter-efficient adaptation with a dual-query grounding strategy.
- **Findings**: The vision-language model achieved the highest F1 of the five (0.911), with strong semantic classification, but weaker localisation precision than the specialised detectors and a latency that rules it out for real-time use. Separately, an ablation showed models pre-trained on general documents performed worse than equivalent models without that pre-training: prior knowledge from one kind of data can hurt rather than help on a structurally different kind.
- **Open questions**: Quantifying how far these results transfer beyond a single engineering domain, through zero-shot testing of domain shift and few-shot measurement of how quickly a model adapts.

Professional Experience

Buro Happold — Sep. 2025 – Present

Data Science Analyst — *London, UK*

- Fine-tuned and evaluated vision-language and language models (PyTorch, LoRA, SWIFT; Azure ML, Hugging Face APIs) to extract structured information from technical documents; owned the labelling framework and evaluation protocol. Contributed to the peer-reviewed EC3 2026 paper above.
- Built a geospatial risk-scoring system combining five years of time-series outage data with satellite-derived features, tuning the precision/recall operating point against the asymmetric cost of false positives versus missed failures.
- Developed model-validation pipelines in Python on Azure, with explicit checks for robustness to distribution shift.
- AI Super User: guide adoption of AI tooling across the business, running internal sessions and advising colleagues from non-technical disciplines.

Green Place Assets — Jun. 2024 – Jul. 2025

Data Science Intern — *Remote*

- Designed a building-performance index combining time-series volatility, peer benchmarking and multivariate scoring; built gradient-boosting and random-forest models on tabular and geospatial data (out-of-sample $R^2 = 0.81$), validated via cross-validation.

Self-employed Jul. 2018 – Jun. 2024
Mathematics & Statistics Tutor *UK*

- Taught GCSE and A-Level mathematics and statistics over six years, developing the ability to explain technical material to people meeting it for the first time; a skill that carries directly into research writing and cross-disciplinary collaboration.

EDUCATION

University of Bath Sep. 2024 – Jan. 2026
MSc Data Science — Distinction (overall 80.7%) *Bath, UK*

- Statistics 94%, Machine Learning 91%, Bayesian Machine Learning 88%, Research 81%. Training in statistical inference, probabilistic modelling, time-series analysis, deep learning and reinforcement learning.
- Dissertation supervised by Prof. Michael Tipping (see Publications & Research).

University of Manchester Sep. 2021 – Jun. 2024
BSc Mathematics — 2:1 *Manchester, UK*

- Probability theory, mathematical statistics, stochastic processes, graph theory, numerical methods, linear algebra; scientific computing in Python and MATLAB.

COLLABORATION & COMMUNICATION

- Co-authored a peer-reviewed conference paper as part of a multi-disciplinary team, working with facade engineering domain experts to define the annotation ontology and cross-check labelling.
- Played football competitively throughout my life in China, across Europe and in Africa, consistently in team environments where reliability to others matters more than individual ability; six years teaching mathematics and statistics alongside my studies.
- Delivered a collaborative reinforcement learning project (PPO agent) as part of a student team during the MSc.

TECHNICAL SKILLS

- **Programming & tools**: Python (NumPy, SciPy, scikit-learn, statsmodels, PyTorch, JAX, Hugging Face), REST/model API integration, SQL, MATLAB, Git/GitHub, Azure ML, LaTeX.